

Average Trap Classroom Run Kit

TOK Cognitive Science Activity Bank • Mathematics • 25-35 min

Category	Mathematics	Time	25-35 min
Difficulty	Easy	ID	average-trap

Big idea: An average is a choice of representation, not a neutral summary of reality.

One-Minute Teacher Brief

Students compare mean, median, and mode for skewed datasets and evaluate which average supports the most responsible claim.

Student Prompt

Inspect Average Trap Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Average Trap Stimulus A	Student-facing stimulus 1: Revision hours: 1, 1, 2, 2, 3, 3, 18. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Average Trap Stimulus B	Student-facing stimulus 2: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Average Trap Reveal / Twist	Student-facing stimulus 3: Screen time claim: mean looks high, median looks moderate, mode shows the most common routine. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — An average is a choice of representation, not a neutral summary of reality.

Actual Lesson Questions

#	Question for students	Teacher key / cue
1	After inspecting Average Trap Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Average Trap Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.

#	Question for students	Teacher key / cue
6	How does choosing a summary statistic shape knowledge?	Teacher cue: students should move from the classroom example to a general TOK issue.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how an average is a choice of representation, not a neutral summary of reality.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Student Instructions

- Work silently for the first response so you can notice your own starting point.
- Record one knowledge claim, the evidence behind it, and your confidence level.
- After the reveal or comparison, update your claim in a different colour or column.
- Use the debrief to answer: How does choosing a summary statistic shape knowledge?

Setup Checklist

- Choose skewed datasets where one large value pulls the mean away from most cases.
- Ask students which number they would put in a headline before calculating all averages.

Example Stimuli or Materials

- Revision hours: 1, 1, 2, 2, 3, 3, 18.
- Screen time claim: mean looks high, median looks moderate, mode shows the most common routine.

Resources To Use

- Resource pack: <resources/average-trap/index.html>
- Card deck CSV: <resources/average-trap/cards.csv>
- Visual prompt: <resources/average-trap/visual-prompt.svg>
- Average comparison sheet: mean, median, mode, purpose, audience, and possible misuse.
- Headline cards using each average to support a different conclusion.

Run Sequence

1. Give students a skewed dataset and ask them to describe a typical value.
2. Groups calculate or identify the mean, median, and mode.
3. Students match each average to a possible headline or policy claim.
4. Groups decide which average is most responsible for the audience and purpose.
5. Debrief why one correct number can create a partial picture.

Debrief Moves

- Knowledge question: When does a mathematical summary become misleading?
- Who decides which average is most relevant?
- Can a number be accurate but unfair?

Teacher Quick Script

- All three averages can be correct; the TOK question is which one responsibly represents the situation.
- What does this number make visible, and what does it hide?

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...